

A Novel Retrospective-Reading Model for Detecting Chinese Sarcasm Comments of Online Social Network

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Abstract—Through the use of sarcastic sentences on social media, people can express their strong emotions. Therefore, the detection of sarcasm in social media has received more and more attention over the past years. Classifying a sentence as sarcastic or nonsarcastic heavily relies on the contextual information of the sentence. However, only focusing on the features of target text is the main solution of most existing research. Moreover, the scale of publicly available Chinese sarcasm dataset is very small and does not contain the contextual information. To address the issues mentioned above, we build a Chinese sarcasm dataset from Bilibili, which is one of the most widely used social network platforms in China and has a significant number of sarcastic comments and contextual information. As far as we know, our dataset is the first publicly available large-scale Chinese sarcasm dataset including contextual information. Additionally, we have proposed a novel retrospective reading method for detecting sarcasm that leverages contextual information to improve model's performance. The experimental results show the effectiveness of the proposed model and the significance of contextual information for Chinese sarcasm detection: achieving the highest F-score of 0.6942, outperforming existing state-of-the-art (SOTA) approaches. The study presented in this article offers approaches and ideas for future Chinese sarcasm detection studies.

Index Terms—Deep learning, intensive reading, sarcasm detection, sketchy reading.

I. INTRODUCTION

ACCORDING to the Merriam-Webster Dictionary, sarcasm is a form of sarcastic humor that utilizes biting, caustic, and sarcastic language, often aimed at specific individuals.¹ The prevalence of sarcasm on social media platforms, such as Twitter, Facebook, Weibo, and Bilibili, is due to its ability to help people express strong emotions. However, the use of sarcasm poses a challenge for sarcasm classification systems, because sarcastic sentences in an positive or neutral context may convey a negative emotion. Therefore, the accurate detection of sarcastic texts is crucial to avoid the incorrect prediction of sentiment.

A. Challenges

There is some research on sarcasm detection for English [1], Portuguese [2], Dutch [3], Italian [4], and other languages [5], [6]. However, only a limited amount of research on Chinese sarcasm detection has been conducted so far. Currently, research on Chinese sarcasm detection faces two challenges.

- 1) *The first challenge is that the lack of a large-scale Chinese annotation corpus*: Specifically, there is no available Chinese annotation corpus with contextual information in this research area (note that the only dataset with contextual information collected from Guanchazhe [7] was not released). Most of the existing studies are done on small-scale labeled Chinese sarcasm dataset without contextual information [8]. The lack of a large-scale Chinese annotation corpus poses the barrier for Chinese sarcasm detection [9].
- 2) *The second challenge is that the classical detection approaches [8], [10] do not take advantage of the contextual information to extract features, which makes them not perform quite well in detecting Chinese sarcastic texts*: Because the contextual information involves the knowledge of commonsense and background [11], we need to propose the methods that integrating contextual information.

¹<https://www.merriam-webster.com/dictionary/sarcasm>

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B. Contribution and Organization

In this article, we construct and release a benchmark dataset on the Github website² and propose a novel Chinese sarcasm detection model based on the retrospective reader.

The following is a summary of our main contributions

- 1) *In this study, we build a dataset of Chinese sarcastic comments with contextual information and share it on the GitHub website:* Specifically, we first collect 108 641 comments from Bilibili. After manually labeling, we construct a dataset with contextual information, containing 76 231 nonsarcastic texts and 2814 sarcastic texts. Title, introduction, response, and other contextual information are included. *As far as we know, our dataset is the first publicly available large-scale Chinese sarcasm dataset that includes contextual information.* Additionally, this dataset provides a valuable resource for future research in Chinese sarcasm detection and related fields. It sets a new standard for the creation and utilization of sarcasm datasets, facilitating the development of more advanced models.
- 2) *Inspired by the idea of retrospective reader, we present a novel retrospective-reading feature extraction approach for Chinese sarcasm detection, which is the first attempt in this research field:* On the one hand, the proposed model constructed by the feature extraction method based on the retrospective-reading achieves the highest F-score of 0.6942, outperforming existing SOTA approaches; On the other hand, according to the representation learning study, the proposed retrospective-reading network achieved the most discriminating satirical text representation, outperforming other SOTA models, which also proves the effectiveness of the feature extraction approach proposed in this article. This innovative method leverages contextual information, enhancing the model's understanding and detection capabilities. It represents a significant advancement in the field, offering a new perspective on how sarcasm can be effectively identified in complex linguistic environments.
- 3) *A novel retrospective-reading model for detecting Chinese sarcasm comments of online social network is built, which outperforms the widely used SOTA detection approaches according to the evaluation results:* On both balanced and imbalanced dataset, our model outperformed other existing SOTA methods and achieved the best performance. In feature ablation experiments, we illustrate the significance of contextual information in sarcasm detection in Chinese. In component analysis, the importance of Sketchy Reading module and Intensive Reading module is proved. Our extensive evaluation demonstrates the robustness and adaptability of our model across different dataset conditions. These two modules provide a comprehensive approach, capturing both superficial and deep contextual information, thereby enhancing detection accuracy.

Here, we note that a shorter conference version of this article has been published [12]. In this journal version, we have improved and extended much of our work. Specifically, the differences are as follows.

- 1) *The new Chinese sarcasm dataset based on the conference version has been publicly available. we expand the features of contextual information, which includes user information and numerical information, such as the number of likes.*
- 2) *A novel Chinese sarcasm detection framework has been built based on the conference version.*
- 3) *Extensive experiments and analyses have been further undertaken.*
 - a) *We add various performance comparison with the traditional machine learning approaches. (see Section V-A)*
 - b) *We add a Feature Ablation Test of contextual information on the detection method using contextual information (our model, HAN and RCNN-RoBERTa). (see Section V-D)*
 - c) *We add a Comparison Experiment on a new Chinese dataset called Ciron to further evaluate the generalization performance of our model. (see Section V-F)*
 - d) *We add the Representation Learning Study to examine the quality of representation learning. (see Section V-G)*
- 4) *More detailed analysis and explanation has been added.*

The following chapters of this article are arranged in the following order: In Section II, we introduce and summarize related works and developments in the sarcasm detection areas. Section III provides a detailed explanation of the proposed framework, while Section IV introduces a critical assessment of various features of our model. Section V conducts a serious of performance comparison experiments with several classification models (including our proposed model). Finally, in Section VI, we sum up the work we have done and explore future directions for research.

II. RELATED WORK

In the early 21st century, research on text sentiment analysis emerged, and since then, with the growth of Internet related industries, the technology related to text sentiment analysis has continued to advance [13], [14], [15], [16], [17], [18], [19]. As the same time, the development of sentiment analysis technology has led to an increased research in the area of sarcasm detection in recent years. Thanks to the development of computational linguistics and natural language processing techniques, which have enabled the exploration of new ways for identifying and analyzing sarcasm in text-based communication. Consequently, there has been a notable surge in research activity aimed at developing robust and effective approaches for the detection and interpretation of sarcasm in textual data.

In this section, we present a comprehensive summary of recent research on sarcasm detection, which spans a range of theoretical and methodological approaches. This summary is

²<https://github.com/zhsatttem/dataset>

TABLE I
COMPARISON OF DIFFERENT APPROACHES FOR SARCASM DETECTION

Approach	References	Strengths	Weaknesses
Rule-based approaches	Kreuz et al. [20] Riloff et al. [21] Bharti et al. [22]	Straightforward to implement and interpret	Rigid and inflexible limited generalization
Machine learning-based approaches	González-Ibáñez et al. [2] Reyes et al. [23] Touahri et al. [24]	Simplicity and efficiency accessible and quick to implement	Dependence on manual feature extraction, limited in capturing complex linguistic nuances
Deep learning-based approaches	Poria et al. [26] Ghosh et al. [38] Tay et al. [29] Bamman et al. [37] Zhang et al. [42] Potamias et al. [41] Schifanella et al. [44] Cai et al. [46]	Automatic feature extraction, superior performance in understanding intricate patterns	High computational resource demands
Chinese sarcasm detection approaches	Tang et al. [51] Jia et al. [10] Wen et al. [52]	Leverage language-specific insights, enhanced accuracy and contextual understanding	Limited availability of high-quality annotated datasets

intended to provide readers with a thorough understanding of the SOTA in sarcasm detection research and to highlight the ongoing efforts to develop more accurate and reliable detection models in the field of sarcasm detection. We provide a clearer understanding of the strengths and weaknesses of each type of approach, as shown in Table I.

A. Rule-Based Approaches

The early research on sarcasm detection is done based on certain rules. Kreuz et al. [20] tried to detect sarcasm by using linguistic features, and found that some specific words such as parenthesis can enhance the detection approach's performance. As sarcasm is often associated with an implied contrast or disparity between conveyed feeling and user's situation. Riloff et al. [21] found that when positive sentiment words appear in negative sentiment contexts, the sentences are more likely to be sarcastic sentences. Based on this observation, they proposed a rule-based sarcasm detection classifier to find positive sentiment words and their associated negative sentiment contexts in the target text, and they proposed negative sentiment contexts by an iterative algorithm. As the times have evolved, researchers have turned their attention to social media, represented by Twitter. Bharti et al. [22] proposed two rule-based methods to detect sarcastic tweets. The first rule is that if a negative expression occurs in a positive sentence, the sentence would be called "ironic." The second rule used exclamation points when detecting sarcastic texts. The rule-based method can achieve good performance when words or patterns satisfying certain rules in a sentence. However, sarcasm is an implicit impression, which means that it is difficult to detect by rule-based methods. To better capture useful information, researchers try to automatic extract information from data and attempt feature-engineering-based machine learning methods.

B. Classical Machine Learning Approaches

The feature-engineering based methods extract the features of origin data and use them to detect sarcasm [2], [23], [24],

[25]. González-Ibáñez et al. [2] extracted features such as uni-grams, presence and frequency of dictionary-based factors, and the SVM and logistic regression were used these features to detect sarcasm. Reyes et al. [23] constructed some conceptual features, such as emotional scenarios, signatures and style. Then they used some classical machine learning approaches to predict the label of the sentence between politics, irony, humor and education. Touahri et al. [24] use a sentiment analysis system by sarcastic features combined with classical machine learning methods, implementing a multilingual detection model.

Methods based on traditional machine learning always outperform those based on rules. However, these methods need complex manual features, which is difficult and time-consuming to undertake. To solve the problems, most recent methods use neural networks to automatically extract features.

C. Deep Learning Approaches

By reviewing existing approaches using deep learning-based models on sarcasm detection, we categorize the related works into two main groups, which includes target text-based approaches [26], [27], [28], [29], [30], [31], [32], [33], [34], context-based approaches [35], [36], [37], [38], [39], [40], [41]. The target text-based approaches focus on extracting the incongruity patterns in target text. Poria et al. [26] was the first one to use neural network to detect sarcasm by the pretrained convolutional neural network (CNN). Ghosh et al. [27] presented a deep neural network that made full use of CNN, LSTM, and DNN. By contrasting deep learning techniques with support vector machines (SVM), they were able to demonstrate that deep learning methods are more effective in sarcasm detection than conventional machine learning techniques. Attention mechanism is also used and the efficiency in sarcasm detection of attention mechanism is shown. To recognize sarcasm, Tay et al. [29] put forth an attention-based model. First, for the purpose of capturing the disparity and contradiction of a sentence, a multidimensional intraattention mechanism was used by them. Second, they used LSTM to encode the input sequence, and combined the two parts to get final result. The proposed model

outperforms than CNN, LSTM, and DNN. However, sarcasm also depends on contextual information, and if the researcher only pay attention to the target text, it can be very challenging to figure out what the speaker really means.

Recently, researchers increasingly drawing attention to contextual information in the domain of sarcasm detection [35], [36], [37], [38], [39], [40], [41]. The importance of contextual information was first presented in [35]. They demonstrated that the contextual information is important in sarcasm detection. They also proposed a method to detect sarcasm by extracting emotions from contextual information such as comments and forum types [36]. There are some different contextual information used for sarcasm detection. Bamman et al. [37] make use of human-engineered author, audience and response to promote sarcasm detection. Zhang et al. [42] focus on sarcasm detection in conversational scenarios, where current approaches ignore the ambiguity and uncertainty of human language in emotional expression and understanding, and they propose a neural network based on mathematical formalisms of quantum theory and fuzzy logic, where they treat the target utterance to be recognized as a quantum superposition of a set of independent words.

Ghosh et al. [38] highlighted the significance of using conversational contextual information in social media discussions. To model the contextual information and responses of the conversation, they employed various LSTM network types. More recently, researchers try to use pretrained language model to detect sarcasm. Babanejad et al. [39] takes advantage of Bi-LSTM and attention mechanism to extract contextual features, then use BERT [43] to handle the features for sarcasm detection. For the purpose of detecting sarcasm in a given response and its contextual information, Srivastava et al. [40] use CNNs and LSTMs to get a final representation of all the sentences in a contextual information, to determine whether or not the target text is satirical, the representation is then mixed with it. Potamias et al. [41] were the first to recognize various forms of figurative language using an unsupervised pretrained transformer approach. To gather contextual information and detect sarcasm, they integrated pretrained RoBERTa weights with RCNN, and proved that using external knowledge such as linguistics can enhance the perform of pretrained language models.

As social media has grown in popularity recently, contextual information is no longer limited to text, more and more multimodal information appears [44], [45], [46], [47], [48]. Schifanella et al. [44] were the first that concatenated textual and visual information in sarcasm detection. They used a model trained on ImageNet and the Visual semantic models from Yahoo Flickr Creative Commons 100M (YFCC100M) to get different features of the images. They also used features found in the literature (e.g., lexical and subjectivity) and features extracted by a two layer perception based on unigram only. Finally, they simply concatenate features of texts and images to make prediction. Cai et al. [46] extended the previous work [44]. They used three different input modalities: text features, image features, and image attributes. A Bi-LSTM was initialized by attribute features and was then used to extract text features. The modality fusion layer performs a weighted average to

the vectors and pumps them to the classification layer to obtain final prediction. By using cross-modality semantic association and contrast, [47] was able to identify sarcastic tweets. They considered that sarcasm is not only the combination of different modalities, but also the relationship between different modalities. The D&R Net was proposed to capture the contextual contrast of multimodal sarcastic tweets and contextual association between image and text. Chauhan et al. [49] used emojis attached to social media comments as a new modality, and they constructed an extended new dataset based on past multimodal datasets, and then proposed a multimodal multitasking deep learning framework for emoji recognition to improve sarcasm detection tasks in multimodal conversation scenarios.

D. Chinese Sarcasm Detection Approaches

Related studies on Chinese sarcasm recognition are still scarce [50]. Tang et al. [51] created a Chinese corpus used for Chinese satirical text research, and mined some of the satirical patterns from it. Jia et al. [10] experimented on a three-way decisions based feature fusion method, which yielded an improvement over traditional one-step classification in irony detection. These methods only focuses on target text, which makes them not perform well in detecting Chinese sarcastic texts. The dataset they constructed is in small-scale and does not contain the content of contextual information. Wen et al. [52] proposed a method combined sememe knowledge and auxiliary information to construct the representation for sarcasm detection. Although they used contextual information as auxiliary information, they used a new module to construct sememe knowledge, which is time-consuming and increase the complexity of the model. And the dataset with contextual information collected from Guanchazhe was not released. Chinese sarcasm detection still lacks the effective application of contextual information and large dataset. In this article, we publish a comprehensive dataset of benchmark Chinese sarcasm that includes contextual information and propose a detection approach based on retrospective reader to efficiently integrate contextual information and target text.

III. DATASET CONSTRUCTION

The lack of Chinese sarcasm annotation corpus greatly hinders sarcasm detection research in Chinese scenarios, as most existing corpus are in English, with only a few available in Chinese [9]. Most of existing studies are done on small-scale labelled Chinese sarcasm dataset without contextual information [8], [51]. Wen et al. [52] used a dataset with contextual information constructed from a Chinese news website, namely Guanchazhe.³ However, their dataset was not publicly available. Therefore, researchers still lack access to a dataset of Chinese sarcasm that includes contextual information.

Similar to YouTube but with more advanced social features is a video sharing website called Bilibili in China. Bilibili is a highly popular social media platform in China with 237 million monthly active users. We discovered that Bilibili contains a

³<https://www.guancha.cn>

lot of satirical information on a variety of particular subjects. In light of this, we decided to create a dataset using Bilibili. The heading, opening statement, response, etc. are all examples of contextual information. The target text and the contextual information are separated into two parts of the raw data.

In this article, we used the balanced dataset proposed by our previous work [12], which includes 2814 sarcastic data and 2814 nonsarcastic data from Bilibili platform.

A. Data Collection Approach

To collect raw data, we developed a crawler specifically for crawling data from the Bilibili website, and then using this crawler we got a lot of data from this platform. The multithreaded web crawler contains four components: proxy middleware, downloader, page parser, and database. *Proxy middleware*: To collect data, the web crawler needs to bypass the anticrawling mechanism of Bilibili, we use the proxy middleware to extract IP address from the proxy IP pool. Then the web crawler can change the IP address in request to proxy IP to increase the speed of the crawler. *Downloader*: Downloader is responsible for downloading HTML. It contains four subcomponents that cooperate with each other, namely engine, URL generator, request queue, and multiple concurrency downloader. *Page parser*: Page parser is the most important module of the crawler. Page parser receives the HTML page and extract data from it. Page parser contains two subcomponents, user information parser and text parser, they are used to process user information and posts, respectively. *Database*: The function of database is to store the data extracted by the page parser. Finally, we use 108 641 comments, including the contextual information collected from Bilibili, to construct the Chinese sarcasm dataset, which includes 76 231 nonsarcastic texts and 2814 sarcastic texts. The contextual information consists of textual features, such as introduction, title, and replies.

B. Data Preprocessing

The raw data consists of many things which need to be processed before moving onto our proposed model. The first step is to remove the useless hashtag in raw data. Some like “#(sarcasm)” and “#(irony)” are useful but things like “#(boring)” are not useful.

Second, we need to remove the emojis and map the emojis to literal meaning. The emojis in the dataset are plenty, and people often use emojis to express their feelings. So we need to construct the emoji mapping dictionary to get the literal meaning of the emojis, which contains the popular emojis labelled as the Chinese meaning in emojis. In addition, Like [7], we also preprocess raw data; specifically, we keep the exclamation mark while removing invalid strings like extra spaces, identifiers and web links.

C. Manually Labeling

There are five independent annotators to complete the manual annotation process to ensure high-equality annotations, each annotator is a postgraduate student, ranging in age from 22

to 25, and they are all native Chinese speakers. We follow the irony identification procedure (IIP) [54] to label the raw data. For sarcastic ratings, we established three fine-grained categories: 0 (sarcastic), 1 (not sarcastic), and 2 (ambiguous). We carefully considered the input from five annotators for the data that had been flagged as ambiguous to improve the quality of the annotation. We will relabel the ambiguous data if more than four annotators divide the data into the same category, and drop it otherwise. We also follow the following rules to improve the annotation quality. First, the annotator should focus more on the inconsistency between the sentence’s literal meaning and user’s true intention. The sentence is likely to be sarcastic if an obvious contradiction appears. Second, if there is no obvious contradiction between the true meaning and the literal meaning, then the annotators need to read the contextual information carefully, comprehend the target text’s true meaning and then reach a conclusion. We then calculated the kappa coefficient [56] to evaluate the agreement between the annotators. In our article, we report a kappa coefficient $K = 0.92$, indicating a high degree of consensus among the annotators. This high kappa value ensures the reliability of our annotations and the overall quality of the dataset.

Table II contrasts our dataset with preexisting sarcasm corpora including source, data scale and obtaining method. Most of existing Chinese sarcasm dataset have a small scale [8], [51] and do not contain contextual information [8], [51], [53]. Gong et al. [7] constructed a dataset with contextual information from a Chinese news website, namely Guanchazhe, but they did not release the dataset. We are the first to publish a comprehensive benchmark dataset for Chinese sarcasm that includes contextual information. Our dataset is collected from 108 641 comments with contextual information from Bilibili, which includes 76 231 nonsarcastic texts and 2814 manually labeled sarcastic texts. The contextual information consists of textual features, such as introduction, title, and replies.

IV. METHODOLOGY

Fig. 1 shows our novel retrospective-reading model for detecting Chinese sarcasm comments, which is consisted by three modules, encoding module, sketchy reading module and intensive reading module. And the meaning of each symbol in this article is detailed in Table III.

First, we use the Encoder to encode the input and get the initial vector representation. After that, we learn to represent the initial impression and intensive impression about the input through the sketchy reading module and intensive reading module. Finally, we feed these representations into classification module for Chinese sarcasm detection. We will then describe each module of our presented model in great detail.

A. The Encoding Module

In Encoding module, we first to capture useful semantic and syntactic information and consider the target text and contextual information to be $D = \{d_1, d_2, \dots, d_m\}$, which represents the input, m is the quantity of input features. All textual words w_j are mapped into embedding vectors x_j . For each input feature

TABLE II
CHINESE SARCASM DATASET COMPARISON

Dataset	Source	Sarcastic	Nonsarcastic	Total	Method	Publicly Available	Contextual Information
Tang et al. [51]	Plurk	950	0	950	Semiautomatic	✓	×
Lin et al. [53]	PTT	17 256	9373	26 629	Semiautomatic	×	×
Gong et al. [7]	Guanchazhe	2486	89 296	91 782	Manual	×	✓
Xiang et al. [8]	Sina Weibo	968	7734	8702	Manual	✓	×
Our dataset	Bilibili	2814	76 231	79 045	Manual	✓	✓

Note: The bold entries indicate that the metric is the best.

TABLE III
GLOSSARY OF NOTATIONS

Notation	Description
D	The input data consists of contextual information and target text
$d_1, d_2, \dots, d_m$	The number of features in input data
$w; x; h$	Textual word; Embedding vector of textual word; Hidden representation
E	Encoder proposed by Li et al. [55]
H^0, H^1, H^2	The representation of input data; Initial impression; Intensive impression
$H_t; H_c; H_{cs}$	Target text information in H ; Contextual information in H ; Summarized contextual information
$W; b$	The weight matrices; The bias
$\alpha; R$	The attention weight; Representation of input feature through attention layer
$\vec{h}; \overleftarrow{h}$	The forward LSTM hidden state; The backward LSTM hidden state
$h^{\text{fl}}; h^{\text{bl}}$	The last forward hidden state; The last backward hidden state
M	The representation of target text information H_t and contextual information H_{cs}
$O; C$	The output vectors in output layer; Features of M through CNN layer

$$R_j = \sum_{i=1}^n \alpha_i h_{ji} \quad (3)$$

where h_{ji} is the i th word embedding of the j th input feature D ; W_1 and W_2 are weight matrices; b_1 and b_2 are biases; the sequence has a length of n . Finally, we get the representation of the initial impression through the multilayer perceptron (MLP)

$$R_j = \text{Attention} \{h_j\} \quad (4)$$

$$H^1 = \{R_1, R_2, \dots, R_m\}. \quad (5)$$

C. The Intensive Reading Module

To obtain the representation of the intensive impression, we propose a module named intensive reading module. To more accurately capture the intensive features, this module extracts features of contextual information and target text respectively by a hierarchical method. We first regard contextual information and target text as different inputs. The Bi-LSTM is used to obtain the temporal and contextual features. In contrast to sketchy reading module, Intensive Reading module focuses more on the target text, so the contextual information is converted into auxiliary information by the summarizer. After that, the auxiliary information is concatenated with the representation of target text and we pass it through the CNN layer to capture the relation

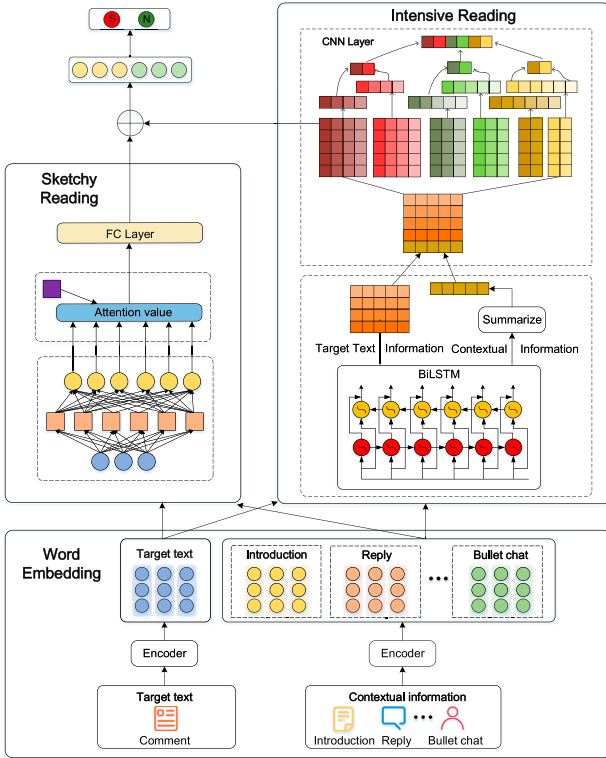


Fig. 1. Proposed Chinese sarcasm detection model.

d_j , the Encoder E proposed by Li et al. [55] converts them into a vector representation h_j , where $h_j = \text{Embedding} \{d_j, E\}$.

To obtain the input vector representation $H^0 = \{h_1, h_2, \dots, h_m\}$, we use the pretrained dense Chinese word vector proposed by Li et al. [55].

B. The Sketchy Reading Module

In this section, we design the sketchy reading module to capture input's preliminary impression. Furthermore, we stitch together the feature representations of the target text and contextual information, then consider it as the input of the sketchy reading module and utilize attention mechanism to get an preliminary impression of the sarcasm detection problem. The attention layer tends to extract the crucial information and enables the model focus on it. Following are the relevant equations:

$$\alpha_i = W_2 \cdot \tanh(W_1 \cdot h_{ji} + b_1) + b_2 \quad (1)$$

$$\alpha = \text{softmax}(\alpha) \quad (2)$$

between the two parts. Finally, we are able to represent the intense impression.

1) *Bi-LSTM Encoder Layer*: The input vector after encoding is divided into contextual information and target text. For each text of the input, a hidden representation vector is obtained by utilizing the Bi-LSTM network to transform a textual sequence. the following equations describe the actions taken by the LSTM:

$$i_t = \sigma(W_i \cdot x_t + U_i \cdot h_{t-1} + b^i) \quad (6)$$

$$f_t = \sigma(W_f \cdot x_t + U_f \cdot h_{t-1} + b^f) \quad (7)$$

$$o_t = \sigma(W_o \cdot x_t + U_o \cdot h_{t-1} + b^o) \quad (8)$$

$$\tilde{c}_t = \tanh(W_c \cdot x_t + U_c \cdot h_{t-1} + b^c) \quad (9)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (10)$$

$$h_t = o_t \odot \tanh(c_t) \quad (11)$$

where W^* , U^* , and b^* are weight matrices which are learned during the process; the $\odot$ denotes element-wise product and the operator σ is the sigmoid function; x_t is the input state after encoding in time-step t , h_t is the final hidden state at time step t . We use these hidden states to integrate additional information into the intensive reading module. For each word in a sequence, we get the forward and backward hidden state $h_t = [\vec{h}_t, \overleftarrow{h}_t]$. Then we can get the new representations of target text $H_t = \text{BiLSTM}(H_t^0)$ and contextual information $H_c = \text{BiLSTM}(H_c^0)$.

2) *Summarization Layer*: The intensive reading module's goal is to focus more attention on the target text, so we need to summarize the contextual information to avoid the large size contextual information vectors to be processed and make the intensive reading module better focus on the target text. For instance, if a sentence can have a maximum length of 100 characters and there are 10 responses in the contextual information, the final vector will have a size of $13 \times 100 \times d_{lstm}$. The forward and backward hidden states of all words from target text H_t are reserved, because we want to save information of the target text as much as possible. In this layer, we extract the last forward hidden state and the last backward hidden state of the input feature i th to represent the summarized contextual information $H_{cs} = \{h_1, h_2, \dots, h_{m-1}\}$, where $h_i = \{h_i^f, h_i^b\}$. $H_{cs} \in \mathbb{R}^{(m-1) \times d_{lstm}}$ is the ultimate output dimension.

3) *CNN Layer*: The connection between the neighboring contextual information and the target text must be recorded in this layer. We first concatenate the representation of target text and auxiliary information summarized by contextual information. The hybrid multichannel CNN is designed to capture N-grams features. Distinct 1-D convolutional layers are employed to capture diverse associations between auxiliary information and the target text. We concatenate the H_t and H_{cs} as input M

$$c_i = f(W \cdot M_{i:i+h-1} + b) \quad (12)$$

$M_{i:i+h-1}$ is the all subsequent $h-1$ parts and the i th part of M are concatenated. $W \in \mathbb{R}^{h \times d_{lstm}}$ is the filter which is learned during the training phase. h is the window size which means the number of features of M used in the process. b is bias and f means the function. The representation M is subjected to a

filter to create a feature map $C = \{c_1, c_2, \dots, c_{m+n-h}\}$. And we can get the most relevant feature $\hat{c}$ after the MAXPOOL layer. The intensive impression H_2 of the input data consists of these most relevant features. And we concatenate the intensive impression H_2 and initial impression H_1 as final representation O_1 of the input data.

D. The Output Layer

For final prediction, We first use a Dropout layer to avoid over-fitting, and feed O_2 into a dense layer with ReLU function. Finally, We get the final prediction through the fully connected layer and softmax function

$$O_2 = \text{Dropout}(O_1) \quad (13)$$

$$O_3 = \max(0, W_2 \cdot O_2 + b_2) \quad (14)$$

$$O_f = \text{softmax}(W_3 \cdot O_3 + b_3) \quad (15)$$

where W^* represents the weight matrix and b^* represents the bias.

V. EXPERIMENTS AND ANALYSIS

The effectiveness of the presented model for sarcasm detection is demonstrated in this section. All experiments were conducted on the workstation, which has an Intel Core i9-10900 processors and NVIDIA GeForce RTX 3070 GPUs with 8 GB of RAM. In the article, we use precision (Pre), recall (Rec), accuracy (Acc), and F-score to measure the performance of the sarcasm detection approaches.

A. Performance Comparison With Traditional Machine Learning Methods

Within the section, we performed an exhaustive comparison with some classic machine learning approaches, including logistic regression (LR), support vector machine (SVM), decision tree (DT), random forest (RF), eXtreme gradient boosting (XGB), naive Bayes, light gradient boosting machine (LightGBM), adaptive boosting (AdaBoost), and they are briefly described as follows:

- 1) *LR*: LR is a widely used method for solving classification problems, including sarcasm detection. Due to the simplicity and interpretability, it's popular among nature language processing (NLP) researchers. LR aims to estimate the probability of a binary output variable by using a logistic function. This function is optimized by maximizing the likelihood of the observed data. This makes logistic regression models well-suited for training on large dataset to accurately classify sarcastic statements. Despite the challenges of sarcasm detection, LR has proven to be effective in accurately classifying sarcastic text [37].
- 2) *SVM*: It is a powerful machine learning algorithm that has been widely used for various classification tasks. In sarcasm detection, SVM have been used with great success to classify sarcastic and nonsarcastic text. One of the advantages of SVM is that SVM can handle high-dimensional feature spaces, making it suitable for NLP tasks that often involve large feature sets [26].

TABLE IV
PERFORMANCE OF OUR MODEL AND DIFFERENT MACHINE LEARNING MODELS ON A BALANCED DATASET

Model	Acc	Sarcasm			Nonsarcasm		
		Pre	Rec	F-score	Pre	Rec	F-score
XGB [57]	0.5940	0.5853	0.5285	0.5554	0.6006	0.6544	0.6264
DT [23]	0.5575	0.5690	0.3213	0.4107	0.5532	0.7754	0.6457
SVM [26]	0.5666	0.7143	0.1616	0.2636	0.5486	0.9404	0.6930
LR [37]	0.5739	0.6746	0.2167	0.3281	0.5556	0.9035	0.6880
RF [58]	0.5547	0.6610	0.1483	0.2422	0.5419	0.9298	0.6848
Naive Bayes [59]	0.5912	0.6757	0.2852	0.4011	0.5698	0.8737	0.6898
LightGBM [60]	0.5812	0.6218	0.3251	0.4270	0.5676	0.8175	0.6700
AdaBoost [61]	0.5575	0.5798	0.2833	0.3806	0.5507	0.8105	0.6558
Our model	0.6953	0.6764	0.6897	0.6635	0.7121	0.7000	0.7246

Note: The bold entries indicate that the metric is the best.

- 3) *DT*: It is a popular machine learning algorithm that can be used for text classification tasks, including sarcasm detection. DT works by dividing the input feature space into smaller subsets based on rules that improve the classification accuracy. This process is repeated recursively until a stopping criterion is met. The rules are typically based on the values of individual features or combinations of features [23].
- 4) *RF*: RF [58] is a popular ensemble learning method that has gained significant attention in the machine learning community due to their high performance, flexibility, and ease of use. RF has been widely used in various classification tasks, including sarcasm detection.
- 5) *XGB*: The XGBoost algorithm proposed by Chen and Guestrin [57], which reduce the loss by continuously adding new trees and fitting residuals. XGB has an excellent performance in text classification.
- 6) *Naive Bayes*: A classical machine learning algorithm based on Bayes theorem, has been extensively used in various classification tasks, including sarcasm detection. Due to the simplicity, efficiency, and ability to handle high-dimensional feature spaces, Naive Bayes has demonstrated competitive performance in sarcasm detection [59].
- 7) *LightGBM*: LightGBM is a novel gradient boosting decision tree proposed by Qi [60], which improves some shortcomings in XGB and has been used in many different kinds of NLP tasks.
- 8) *AdaBoost*: Adaboost [61] is an ensemble learning algorithm used in machine learning for classification problems, and has been widely used for sarcasm detection. It works by combining multiple “weak” classifiers into a single “strong” classifier.

Considering the limitations of traditional machine learning approaches in input data structure, we only provide the target text to the relevant machine learning models. The comparison results of the four traditional machine learning approaches for detecting sarcastic text are provided in Table IV.

Based on the information presented in the table and figures, it is evident that the performance of our model is optimal, indicating the effectiveness of the retrospective reader. Our model’s superior performance can be attributed to its ability to accurately identify and analyze sarcasm in text-based communication, a

crucial aspect of sentiment analysis. Among the five machine learning algorithms, the best performing algorithm is XGB. This superiority can be attributed to XGB’s strong robustness and noise immunity. This is because XGB uses a combination of gradient boosting and regularization techniques, which helps prevent overfitting and enhances its ability to generalize well to new data. RF also shows good performance by building multiple decision trees, and DT gets lower performance than RF. However, the simple machine learning approaches of LR and SVM have almost no advantages over the ensemble learning approaches of RF and XGB. Upon analysis, it has been determined that the inferior performance of LR and SVM compared to ensemble learning methods, such as RF and XGB, is attributed to their inability to effectively handle high-dimensional sparse data, susceptibility to noise and outliers, and limited ability to address nonlinear relationships in text classification tasks. Given that text classification data often involves high-dimensional and sparse data, the performance of LR and SVM can be hindered. Furthermore, the linearity of LR and SVM hampers their capacity to accurately capture complex nonlinear relationships and interaction terms present in text classification tasks.

Recently, researchers utilize the powerful techniques of neural networks to get more precise semantic representations of sarcastic text and model the sequential information of sarcastic contextual information. In other experiments, we will focus on the techniques of neural networks.

B. Performance Comparison With Deep Learning Methods

Within this section, we implemented the following baselines to serve as a comparison for our proposed model:

- 1) *TextCNN*: One of the most advanced neural network-based approaches, has emerged as a powerful method for solving text classification problems. TextCNN is a convolutional neural network. It first converts the text into an embedding matrix and then applies convolutional kernels of different sizes to this matrix. Finally, it extracts the maximum or average values of each kernel through pooling operations [53].
- 2) *DPCNN*: TextCNN finds it challenging to convey long-distance associations in text using convolution, whereas DPCNN [64] can do so by deepening the network.

TABLE V
SARCASM DETECTION RESULTS OF DIFFERENT DEEP LEARNING MODELS

Model	Acc	Sarcasm			Nonsarcasm		
		F-score	Pre	Rec	F-score	Pre	Rec
LSTM [62]	0.6211	0.6305	0.5943	0.6714	0.6112	0.6531	0.5744
LSTM+CNN [63]	0.6475	0.6282	0.6263	0.6302	0.6648	0.6667	0.6630
LSTM+attention [62]	0.6416	0.5963	0.6514	0.5479	0.6778	0.6349	0.7269
TextCNN [63]	0.6484	0.6211	0.6399	0.6033	0.6721	0.6554	0.6897
DPCNN [64]	0.6143	0.4968	0.3955	0.6678	0.6873	0.5929	0.8173
HAN [65]	0.6578	0.6425	0.6444	0.6407	0.6719	0.6702	0.6737
ERNIE [66]	0.6442	0.5548	0.6943	0.4620	0.7036	0.6206	0.8123
BERT [43]	0.6182	0.5982	0.6152	0.5820	0.6363	0.6207	0.6527
RCNN-RoBERTa [67]	0.6615	0.6279	0.6645	0.5951	0.6895	0.6592	0.7228
Our	0.6953	0.6764	0.6897	0.6635	0.7121	0.7000	0.7246

Note: The bold entries indicate that the metric is the best.

- 3) *LSTM*: One of the most common techniques for resolving text classification issues, which can capture the longer dependency in sequence for sarcasm classification [62].
- 4) *LSTM-attention*: We also incorporate the LSTM [62] and attention mechanism [68] to get better performance for sarcasm detection.
- 5) *LSTM-CNN*: CNN [63] is used to extract features in target text, which contains two convolutional layers and a fully connected layer.
- 6) *Hierarchical attention network*: The attention mechanism is used by HAN [65] to categorize documents. We create a document using target text and contextual information, and HAN is used to classify it. With regard to sarcasm recognition, HAN has a strong foundation and is capable of using contextual information [40].
- 7) *BERT*: The BERT [43] is a pretrained language model that makes use of the Transformer's encoder and performs better than approaches that only pay attention to the target text, which is widely used in many NLP tasks.
- 8) *ERNIE*: ERNIE [66] is a popular method for solving Chinese NLP problems, which learned knowledge of longer semantic dependency and knowledge-related information.
- 9) *RCNN-RoBERTa*: The pretrained RoBERTa vectors are used by Potamias et al. [67] and the RCNN to capture contextual information for sarcasm detection.

Table V displays how our model and other deep learning models performed on the dataset. Figs. 2 and 3 display the ROC and PR curves respectively.

From Table V, we can see that Bi-LSTM and TextCNN perform better than DPCNN, which indicates that they can extract features more effectively from comments. The extraction ability of DPCNN for some semantic features is relatively weak, which is due to the fact that the DPCNN model mainly extracts local features by convolution and pooling when dealing with text classification tasks, but the ability of extracting semantic features is relatively weak, while TextCNN and Bi-LSTM are stronger than DPCNN in terms of feature extraction ability. This is likely because TextCNN leverages multiple convolutional filters of different sizes, allowing it to capture a variety of n-grams and semantic patterns more comprehensively. Bi-LSTM, on the other hand, benefits from its ability to maintain information

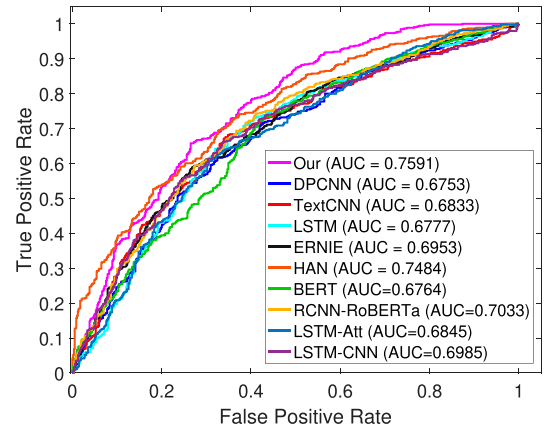


Fig. 2. ROC curve.

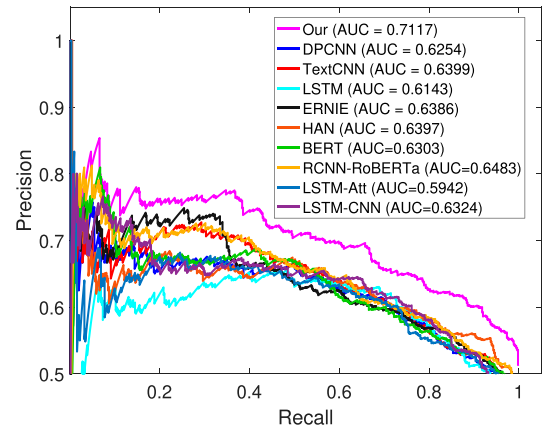


Fig. 3. PR curve.

over long sequences and capture the temporal dependencies within the text. Besides that, LSTM+CNN and LSTM+attention perform better than Bi-LSTM, which indicates that CNN and attention mechanism can improve the performance for detecting Chinese sarcasm comments of online social network. The combination of LSTM and CNN leverages the sequential processing power of LSTM with the feature extraction capabilities of CNN, leading to better feature representation. The attention mechanism further enhances the model by focusing on the most

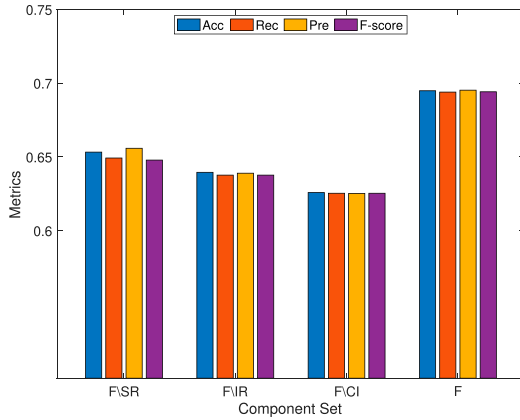


Fig. 4. Sarcasm detection results of different ablated models.

relevant parts of the input sequence, effectively highlighting the key features necessary for accurate sarcasm detection. Additionally, it is evident that methods that utilize contextual information to extract features, such as RCNN-RoBERTa and HAN, outperform those that solely concentrate on the target text, such as DPCNN, Text CNN, and Bi-LSTM. Our model performs more effectively than the pretrained models ERNIE and BERT. The pretraining data field is likely to have an impact on ERNIE and BERT, therefore, it is difficult to apply them perfectly to this domain. Since our model loses part of sentence information during summarizing module, our model did not achieve the best Rec compare to LSTM, DPCNN, and ERNIE. The best Acc and F-score were attained by our model. In Fig. 2, the AUC score between methods that incorporate contextual information and approaches that only focus on target text illustrates the significance of contextual information.

C. Analysis of the Components of the Proposed Model

The analysis of the components allowed us to gain valuable insights into the contribution of each component to the overall performance of the model. These insights can be used to improve the design of future models by determining which components are most critical for optimal performance. Ultimately, the analysis of the components provides a valuable tool for comprehending the inner workings of complex models, as well as improving their effectiveness in various applications.

In this experiment, to elucidate the individual effects of the proposed model's various components, we evaluate the effectiveness of each part of the proposed model, specifically, the sketchy reading module, intensive reading module and contextual information. As a multicomponent model, F encompasses several significant components, including first reading module and intensive reading module, as well as contextual information. To evaluate the performance of our model with and without these components, we systematically removed each of them and conducted experiments to assess the performance of the resulting models. The three models obtained by removing these components were named $F\backslash SR$, $F\backslash IR$, and $F\backslash CI$.

From Fig. 4, we can see that both the retrospective reader and contextual information can improve the performance of our

model. And the Intensive Reading module is more important than sketchy reading module, which indicates that intensive impression can provide more crucial information than initial impression. This suggests that a deeper, more thorough analysis of the text yields better results in detecting sarcasm. Besides that, we can see that the ablated model $F\backslash CI$ performs worst which shows that the contextual information is the most useful part of our model and important for detecting Chinese sarcasm comments of online social network. Including contextual information aids in understanding the surrounding context of a comment, which is often essential for accurately detecting sarcasm. Therefore, it is essential to include contextual information as a component in future models for accurately detecting sarcasm in Chinese online social network comments, additionally, the result also can make a contribution to better understanding and communication in online social networks.

D. Feature Ablation Test

To further illustrate the significance of contextual information in understanding Chinese sarcasm, we conduct feature ablation test on the methods utilizing contextual information (HAN, RCNN-RoBERTa, and the proposed model). Fig. 5 demonstrates the outcomes of the feature ablation experiment.

Following are the features we used for feature ablation:

- 1) H : The proposed model's entire feature set will be applied.
- 2) $H\backslash Keywords$: only includes comment, title, reply.
- 3) $H\backslash Title$: only includes comment, reply, keywords.
- 4) $H\backslash Reply$: only includes comment, keywords, title.
- 5) $H\backslash CI$: only includes comment.

The results indicate that models utilizing contextual information outperform other models on the feature set H that includes all features, and perform the worst on the feature set $H\backslash CI$. This proves that each feature of contextual information is important, and they can enhance the model's performance. When approaches perform on feature set $H\backslash Reply$, their performance is worst among $H\backslash Reply$, $H\backslash Title$ and $H\backslash Keywords$. This suggests that the reply feature has the highest level of distinguishability. In addition, the performance of approaches using feature set $H\backslash Keywords$ is similar to that of H , which indicates that the distinguishability of keywords feature is the smallest. Keywords alone may not provide sufficient context or nuance for detecting sarcasm effectively. Furthermore, the performance of approaches utilizing the feature set $H\backslash Keywords$ is comparable to that of H , suggesting that the distinguishability of the keywords feature is the least significant. The outcomes highlight the significance of contextual information for Chinese sarcasm recognition. Integrating a broader range of contextual features could further enhance the model's ability to detect sarcasm. If we can increase the number of features in contextual information, it may improve the performance of detection models.

E. Robustness Analysis With the Baseline Approaches

This section compares our model to the baselines in realistic scenarios. We use the unbalanced dataset to conduct model comparison experiments to assess the performance of our model

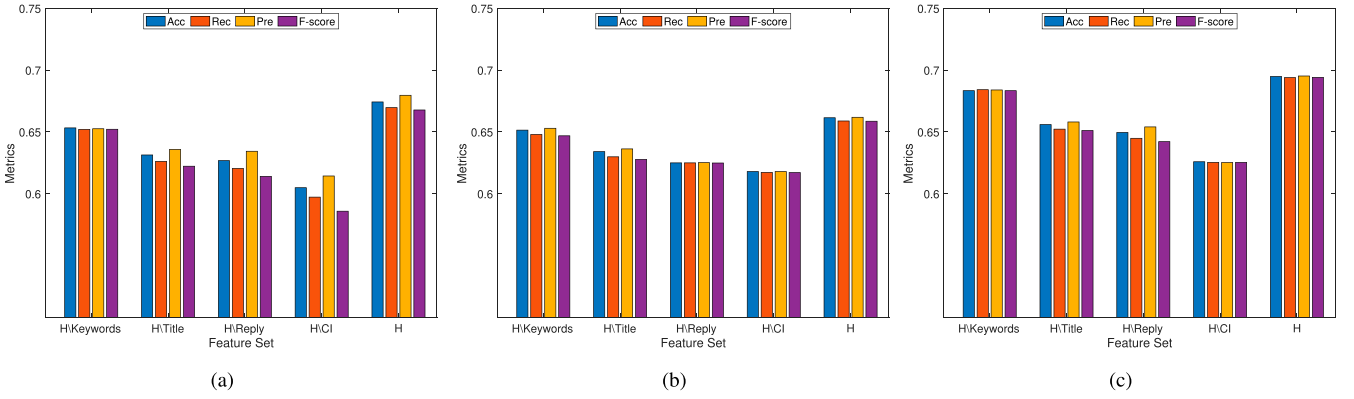


Fig. 5. Results of feature ablation test. (a) HAN in feature ablation test. (b) RCNN-RoBERTa in feature ablation test. (c) Our model in feature ablation test.

in real-world scenarios. We first set the size of our dataset to 3000, and then we continually to increase the proportion of satirical texts to 90% according to the growth rate of 10%. Fig. 6 shows the results.

Our model outperforms the approaches using contextual information (RCNN-RoBERTa and HAN) and the approaches only concentrate on target text (DPCNN, TextCNN, ERNIE, BERT, BiLSTM). The reason for this is that our method, using a retrospective reader, can more effectively capture the connection between the target text and the contextual information. We can also see that the methods using contextual information (our method, RCNN-RoBERTa and HAN) perform better than other methods (BiLSTM, TextCNN, DPCNN, ERNIE, BERT). This means that contextual information is helpful for the model to acquire more knowledge on the input data. The performance of our model and ERNIE are very close when the percentage of sarcastic data exceeds 60%. Additionally, when the percentage of sarcastic data is lower than 60%, our model outperforms ERNIE in terms of F score. This suggests that our model is more effective for detecting sarcastic texts in scenarios where sarcasm is not as prevalent. It also indicates the importance of dataset balance in training sarcasm detection models. Our model's superior performance in less sarcastic data scenarios highlights its robustness and ability to generalize better across varying levels of sarcasm prevalence.

When less than 50% of the data is sarcastic, our model outperforms other approaches as well. The robustness of our model is demonstrated by the fact that it consistently outperforms competing methods at data with varying percentages of sarcasm.

The efficacy of our proposed model in sarcasm detection is demonstrated by the remarkable robustness across dataset with varying percentages of sarcastic data, exhibiting superior performance over competing methods. This impressive performance attests to the model's ability to effectively navigate the intricacies of sarcasm detection across diverse contextual information, making it a reliable tool for precise classification even in challenging scenarios. Additionally, the model's robustness underscores the potential value in practical applications, where dataset may present widely varying proportions of sarcastic data.

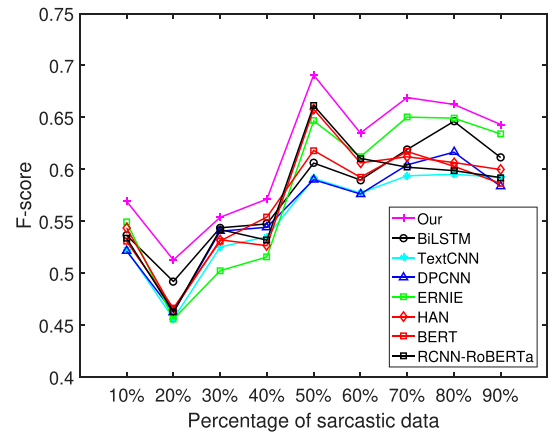


Fig. 6. Robustness comparison of different models.

F. Performance Comparison on Another Chinese Sarcasm Dataset

To further evaluate the real-world performance of our model, we carry out the comparison experiment using the Ciron dataset, a new Chinese dataset. Xiang et al. [8] explored the characteristic features of sarcasm in Chinese and provided an imbalanced Chinese sarcasm dataset, referred as Ciron, which includes 7734 nonsarcastic target text and 968 sarcastic target text. Ciron dataset consists only of target text. Same as before, we split the data into training, validation, and test sets in an 80%:10%:10% ratio.

Table VI shows the experimental results. We can see that the proposed approach has better performance than the approaches of convolutional neural networks (Text CNN, DPCNN), the detection approaches of recurrent neural networks (BiLSTM, BiLSTM+CNN, BiLSTM+Attention) and the detection approach of BERT (BERT, ERNIE). Moreover, compared to the CNN and RNN approaches, BERT detection approach have been improved to a certain extent. According to our analysis, this may be due to the advantages of BERT's stronger contextual information awareness, zero-sample learning ability, multitask learning ability compared to CNN and RNN for text classification tasks, which is why BERT performs better than RNN and CNN in our experiments. This enhanced performance is

TABLE VI
 METRICS OF OUR MODEL AND OTHER DEEP LEARNING MODELS ON THE CIRON DATASET

Model	Acc	Sarcasm			Nonsarcasm		
		F-score	Pre	Rec	F-score	Pre	Rec
DPCNN [64]	0.8895	0.3239	0.5111	0.5371	0.9398	0.9102	0.9715
LSTM [62]	0.8608	0.3459	0.3636	0.3299	0.9221	0.9168	0.9275
TextCNN [63]	0.8930	0.4076	0.5333	0.3299	0.9412	0.9197	0.9637
LSTM+CNN [63]	0.8872	0.4368	0.4936	0.3918	0.9373	0.9255	0.9495
LSTM+Attention [62]	0.8838	0.3727	0.4688	0.3093	0.9360	0.9168	0.9560
ERNIE [66]	0.9091	0.4476	0.6957	0.3299	0.9505	0.9210	0.9819
BERT [43]	0.9010	0.3385	0.6667	0.2268	0.9465	0.9103	0.9858
Our	0.9125	0.4795	0.7143	0.3608	0.9523	0.9244	0.9819

Note: The bold entries indicate that the metric is the best.

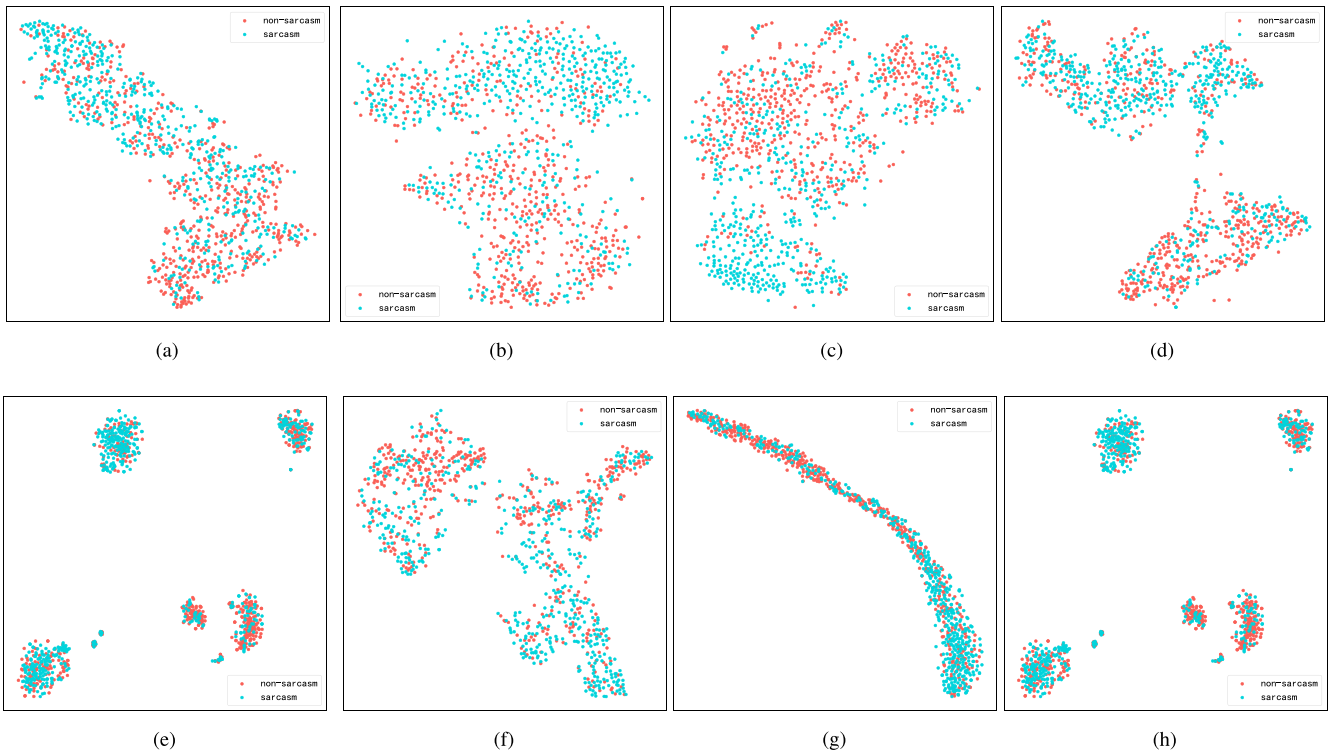


Fig. 7. Plots of text representations. (a) Text representation of HAN. (b) Text representation of RCNN-RoBERTa. (c) Text representation of RR-NET. (d) Text representation of DPCNN. (e) Text representation of TextCNN. (f) Text representation of LSTM. (g) Text representation of LSTM-Att. (h) Text representation of LSTM-CNN.

attributed to BERT's ability to better understand and utilize the context of the input data, making it more effective for tasks requiring deep semantic comprehension. In light of the findings, it can be concluded that the proposed model performs better in real-world scenarios than the baseline approaches. This suggests that our model, which incorporates advanced contextual understanding, is well-suited for practical applications where nuanced text interpretation is crucial.

G. Representation Learning Study

Based on the results presented in this study, the proposed model and two other contextual detection methods achieved the highest performance in the model comparison experiments.

To check the quality of the sarcastic text learned by the model proposed in this article, a comparison was conducted with other deep learning-based baseline models. In this section, we will extract the vector of the input classification layer as model's semantic representation. Using the t-SNE dimensionality reduction technique, the comparison results is shown in Fig. 7, where the red dots are sarcastic data.

The findings reveal that the proposed retrospective-reading network (RR-NET) achieved the highest discrimination, outperforming other baseline models. Compared to other baseline models, DPCNN, HAN and RCNN-RoBERTa all achieve a more discriminative representation of sarcastic text.

Among these baseline models, HAN and RCNN-RoBERTa are able to fuse contextual information with the target text

to obtain more sarcastic semantics and thereby obtain better discrimination.

It is worth noting that DPCNN also obtains good discrimination, but it performs poorly in the model comparison experiments, which may be because the simple fully connected layer cannot capture the sarcasm semantic features provided by DPCNN, and it may be possible to adopt a more excellent feature capture method to improve the performance of the DPCNN-based Chinese sarcasm detection model in future work.

The model proposed in this article has the best discrimination, with a large amount of sarcastic data gathered in the lower left corner of the image and nonsarcastic data distributed above the sarcastic data, which is also consistent with the results in the model comparison experiments. Future research could explore integrating more sophisticated layers or hybrid approaches to enhance the ability of DPCNN to interpret and utilize sarcasm-specific features.

VI. CONCLUSION AND FUTURE WORK

In this article, we have proposed a novel model for sarcasm detection based on retrospective-reading model. The model mainly consists of three modules: data collection and labeling module, feature extraction module, and the detection module. Specifically, a dataset including comments, titles, keywords and replies was constructed to detect sarcastic texts from Bilibili with contextual information. Moreover, by making use of the attention mechanism, the Bi-LSTM, the CNN, and MLP, we designed a new deep neural network model with retrospective reader for Chinese sarcasm detection. According to the experimental findings, Our proposed model performs better than other SOTA detection approaches for sarcastic texts.

In future work, several promising research directions can be explored to advance the field of sarcasm detection. Incorporating large language models for enhanced sarcasm detection. Future research could explore the integration of large language models, such as ChatGPT, to enhance the accuracy and robustness of sarcasm detection systems. These models can capture complex linguistic patterns and contextual nuances, potentially leading to significant improvements in sarcasm detection. Cross-lingual sarcasm detection. While our current work focuses on Chinese sarcasm detection, an interesting direction for future research is to develop cross-lingual sarcasm detection models. These models aim to detect sarcasm across multiple languages, taking into account linguistic nuances and cultural differences. Utilizing knowledge graphs. In future research, based on the extent to which sarcastic expressions rely on external knowledge, knowledge graphs contribute to enhancing the natural language understanding of sarcasm. Therefore, utilizing knowledge graphs might significantly improve the effectiveness of sarcasm detection. By focusing on these three directions, future researchers can build upon our work and contribute to the ongoing development of more sophisticated and effective sarcasm detection systems.

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